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Double single level cell program in a memory device

Abstract

Control logic in a memory device causes a pass voltage to be applied to a plurality of wordlines of a block of a memory array of the memory device, the block comprising a plurality of sub-blocks, and the pass voltage to boost a channel potential of each of the plurality of sub-blocks to a boost voltage. The control logic further selectively discharges the boost voltage from one or more of the plurality of sub-blocks according to a data pattern representing a sequence of bits to be programmed to respective memory cells of the plurality of sub-blocks. In addition, the control logic causes a single programming pulse to be applied to a selected wordline of the plurality of wordlines of the block to program the respective memory cells of the plurality of sub-blocks according to the data pattern.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/308,813, filed Feb. 10, 2022, the entire contents of which are hereby incorporated by reference herein.

TECHNICAL FIELD

(1) Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to double single level cell (SLC) program in a memory device of a memory sub-system.

BACKGROUND

(2) A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure.

- (2) FIG. 1A illustrates an example computing system that includes a memory sub-system in accordance with some embodiments of the present disclosure.
- (3) FIG. 1B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system, in accordance with some embodiments of the present disclosure.
- (4) FIG. 2 is a schematic of portions of an array of memory cells as could be used in a memory of the type described with reference to FIG. 1B in accordance with some embodiments of the present disclosure.
- (5) FIG. 3 is a schematic of portions of an array of memory cells implementing double single level cell (SLC) programming in accordance with some embodiments of the present disclosure.
- (6) FIGS. 4A-4C are signal diagrams illustrating various signals applied to a memory array during a double single level cell (SLC) programming operation in accordance with some embodiments of the present disclosure.
- (7) FIG. 5 is a flow diagram of an example method of double single level cell (SLC) programming in a memory device of a memory sub-system in accordance with some embodiments of the present disclosure.
- (8) FIG. 6 is a block diagram of an example computer system in which embodiments of the present disclosure can operate.

DETAILED DESCRIPTION

- (9) Aspects of the present disclosure are directed to double single level cell (SLC) program in a memory device of a memory sub-system. A memory sub-system can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIG. 1A. In general, a host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.
- (10) A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. For example, NAND memory, such as 3D flash NAND memory, offers storage in the form of compact, high density configurations. A non-volatile memory device is a package of one or more dice, each including one or more planes. For some types of non-volatile memory devices (e.g., NAND memory), each plane includes a set of physical blocks. Each block includes of a set of pages. Each page includes of a set of memory cells (“cells”). A cell is an electronic circuit that stores information. Depending on the cell type, a cell can store one or more bits of binary information, and has various logic states that correlate to the number of bits being stored. The logic states can be represented by binary values, such as “0” and “1”, or combinations of such values.
- (11) A memory device can be made up of bits arranged in a two-dimensional or a three-dimensional grid. Memory cells are etched onto a silicon wafer in an array of columns (also hereinafter referred to as bitlines) and rows (also hereinafter referred to as wordlines). A wordline can refer to one or more rows of memory cells of a memory device that are used with one or more bitlines to generate the address of each of the memory cells. The intersection of a bitline and wordline constitutes the address of the memory cell. A block hereinafter refers to a unit of the memory device used to store data and can include a group of memory cells, a wordline group, a wordline, or individual memory cells. One or more blocks can be grouped together to form separate partitions (e.g., planes) of the memory device in order to allow concurrent operations to take place on each plane.
- (12) During a program operation on a non-volatile memory device, certain phases can be encountered, including program and program verify. For example, a high program voltage can be applied to a selected wordline of a block of the memory device during a program phase, followed by a program verify phase where a verify voltage is applied to the selected wordline. Certain program operations can be single program operations, wherein one sub-block is programmed in

each operation. In such a single program operation, a data pattern is read from a temporary storage location (e.g., a page buffer) to determine whether the memory cell associated with a selected wordline and located in the one sub-block is to be programmed or not, and a single programming pulse can be applied before the program verify phase occurs. This same process can then be repeated for each remaining sub-block to be programmed. Other program operations can be double program operations, for example, wherein two sub-blocks are programmed in one operation. In such a double program operation, the two sub-blocks can be programmed (i.e., two separate programming pulses can be applied) before the program verify phase occurs. Depending on the implementation, certain memory devices can utilize either a double verify operation or a seamless verify operation during the subsequent program verify phase. In either case, programming multiple sub-blocks involves causing multiple separate programming pulses to be applied to the selected wordline. There are latencies associated with each programming pulse including ramping up and down the program voltage multiple times. These latencies increase the length of the program operation, which can be especially impactful in high-priority and time-sensitive operations, such as single level cell (SLC) programming operations.

(13) Aspects of the present disclosure address the above and other deficiencies by implementing double single level cell (SLC) program in a memory device of a memory sub-system. In a double SLC program operation, control logic in the memory device can program memory cells in two or more separate sub-blocks using a single programming pulse applied to the selected wordline. In one embodiment, as part of a programming operation, the control logic causes a pass voltage to be applied to each wordline in a block of the memory device, including the selected wordline (i.e., the wordline associated with the memory cell(s) to be programmed) and unselected wordlines. The pass voltage boosts a memory pillar channel voltage in each sub-block of the memory device to a higher boost voltage during this phase of the program operation. Once each pillar channel voltage is boosted, the control logic can selectively discharge the pillars of one or more sub-blocks according to a data pattern of bits to be programmed to the block during the program operation. For example, if a memory cell associated with the selected wordline and located in a first sub-block is to be programmed, the control logic can activate a select gate device at the drain of that sub-block to allow the boost voltage to be discharged onto the bitline, thereby bringing the channel voltage back to a ground voltage. Conversely, if a memory cell associated with the selected wordline and located in a second sub-block is not to be programmed, the control logic will not activate the select gate device at the drain of that sub-block, thereby causing the pillar channel voltage to remain at the boost voltage. This sequence can be repeated for two or more sub-blocks. Once complete, the control logic can cause a single programming pulse to be applied to the selected wordlines. Those sub-blocks at the ground voltage will be programmed, while those sub-blocks at the boost voltage will be inhibited, thereby allowing multiple sub-blocks to be programmed concurrently via the single programming pulse. Either a double verify operation or a seamless verify operation can then be performed during the subsequent program verify phase.

(14) Advantages of this approach include, but are not limited to, improved performance in the memory device. The double SLC program operation described herein allows for the programming of multiple sub-blocks in SLC memory to be performed concurrently (e.g., simultaneously) using a single programming pulse. This results in fewer program operations being performed (e.g., one half the number of program operations) for the same amount of data being programmed to the memory device. Accordingly, the latency associated with the entire programming operation is reduced which can improve SLC programming performance.

(15) FIG. 1A illustrates an example computing system **100** that includes a memory sub-system **110** in accordance with some embodiments of the present disclosure. The memory sub-system **110** can include media, such as one or more volatile memory devices (e.g., memory device **140**), one or more non-volatile memory devices (e.g., memory device **130**), or a combination of such.

(16) A memory sub-system **110** can be a storage device, a memory module, or a hybrid of a storage

device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) card, and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory modules (NVDIMMs).

(17) The computing system **100** can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things (IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or such computing device that includes memory and a processing device.

(18) The computing system **100** can include a host system **120** that is coupled to one or more memory sub-systems **110**. In some embodiments, the host system **120** is coupled to different types of memory sub-system **110**. FIG. 1A illustrates one example of a host system **120** coupled to one memory sub-system **110**. As used herein, “coupled to” or “coupled with” generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

(19) The host system **120** can include a processor chipset and a software stack executed by the processor chipset. The processor chipset can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller). The host system **120** uses the memory sub-system **110**, for example, to write data to the memory sub-system **110** and read data from the memory sub-system **110**.

(20) The host system **120** can be coupled to the memory sub-system **110** via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate (DDR)), etc. The physical host interface can be used to transmit data between the host system **120** and the memory sub-system **110**. The host system **120** can further utilize an NVM Express (NVMe) interface to access the memory components (e.g., memory devices **130**) when the memory sub-system **110** is coupled with the host system **120** by the PCIe interface. The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system **110** and the host system **120**. FIG. 1A illustrates a memory sub-system **110** as an example. In general, the host system **120** can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

(21) The memory devices **130**, **140** can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device **140**) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

(22) Some examples of non-volatile memory devices (e.g., memory device **130**) include negative-and (NAND) type flash memory and write-in-place memory, such as three-dimensional cross-point (“3D cross-point”) memory. A cross-point array of non-volatile memory can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories, cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously erased. NAND type flash memory includes, for

example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

(23) Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single level cells (SLC) can store one bit per cell. Other types of memory cells, such as multi-level cells (MLCs), triple level cells (TLCs), and quad-level cells (QLCs), can store multiple bits per cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, or a QLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks.

(24) Although non-volatile memory components such as a 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g., 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random-access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, electrically erasable programmable read-only memory (EEPROM).

(25) A memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

(26) The memory sub-system controller **115** can include a processor **117** (e.g., a processing device) configured to execute instructions stored in a local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

(27) In some embodiments, the local memory **119** can include memory registers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. 1A has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system **110** does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

(28) In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., logical block address (LBA), namespace) and a physical address (e.g., physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host

interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

(29) The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

(30) In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An external controller (e.g., memory sub-system controller **115**) can externally manage the memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, a memory device **130** is a managed memory device, which is a raw memory device **130** having control logic (e.g., local controller **135**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device. Memory device **130**, for example, can represent a single die having some control logic (e.g., local media controller **135**) embodied thereon. In some embodiments, one or more components of memory sub-system **110** can be omitted.

(31) In one embodiment, memory sub-system **110** includes a memory interface component **113**. Memory interface component **113** is responsible for handling interactions of memory sub-system controller **115** with the memory devices of memory sub-system **110**, such as memory device **130**. For example, memory interface component **113** can send memory access commands corresponding to requests received from host system **120** to memory device **130**, such as program commands, read commands, or other commands. In addition, memory interface component **113** can receive data from memory device **130**, such as data retrieved in response to a read command or a confirmation that a program command was successfully performed. In some embodiments, the memory sub-system controller **115** includes at least a portion of the memory interface **113**. For example, the memory sub-system controller **115** can include a processor **117** (e.g., a processing device) configured to execute instructions stored in local memory **119** for performing the operations described herein. In some embodiments, the memory interface component **113** is part of the host system **110**, an application, or an operating system.

(32) In one embodiment, memory device **130** includes local media controller **135** and a memory array **104**. As described herein, local media controller **135** can perform a program operation on the memory cells of memory array **104**. A program operation can include, for example, a program phase and a program verify phase. During the program phase, a program voltage is applied to a selected wordline(s) of the memory array **104**, in order to program a certain level(s) of charge to selected memory cells on the wordline(s) representative of a desired value(s). In one embodiment, by conditioning the channel potential associated with multiple sub-blocks according to a data pattern to be programmed to the memory cells contained therein before the program voltage is applied to the selected wordline, multiple memory cells in separate sub-blocks can be accurately programmed using a single programming pulse. For example, at the start of the program operation, local media controller **135** can cause a pass voltage to be applied to a plurality of wordlines of a block of memory array **104** in memory device **130**. The block can include a plurality of sub-blocks, and the pass voltage can boost a channel potential of each of the plurality of sub-blocks to a boost voltage (V_{boost}). Local media controller **135** can further selectively discharge the boost voltage from one or more of the plurality of sub-blocks according to a data pattern representing a sequence of bits to be programmed to respective memory cells of those sub-blocks. This can result in the channel potential of the sub-blocks containing memory cells to be programmed to discharge to a ground voltage. In addition, Local media controller **135** can cause a single programming pulse to

be applied to a selected wordline of the plurality of the block to program the respective memory cells of the plurality of sub-blocks according to the data pattern. In one embodiment, the memory cells in those sub-blocks for which the channel potential was discharged to ground will be programmed, while memory cells in those sub-blocks for which the channel potential was not discharged and remained at the boost voltage will be inhibited from being programmed. A program verify phase can then be initiated to verify that the memory cells were programmed correctly according to the data pattern. Further details with regards to the operations of local media controller **135** are described below.

(33) FIG. **1B** is a simplified block diagram of a first apparatus, in the form of a memory device **130**, in communication with a second apparatus, in the form of a memory sub-system controller **115** of a memory sub-system (e.g., memory sub-system **110** of FIG. **1A**), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller **115** (e.g., a controller external to the memory device **130**), may be a memory controller or other external host device.

(34) Memory device **130** includes an array of memory cells **104** logically arranged in rows and columns. Memory cells of a logical row are typically connected to the same access line (e.g., a wordline) while memory cells of a logical column are typically selectively connected to the same data line (e.g., a bit line). A single access line may be associated with more than one logical row of memory cells and a single data line may be associated with more than one logical column. Memory cells (not shown in FIG. **1B**) of at least a portion of array of memory cells **104** are capable of being programmed to one of at least two target data states.

(35) Row decode circuitry **108** and column decode circuitry **109** are provided to decode address signals. Address signals are received and decoded to access the array of memory cells **104**. Memory device **130** also includes input/output (I/O) control circuitry **160** to manage input of commands, addresses and data to the memory device **130** as well as output of data and status information from the memory device **130**. An address register **114** is in communication with I/O control circuitry **160** and row decode circuitry **108** and column decode circuitry **109** to latch the address signals prior to decoding. A command register **124** is in communication with I/O control circuitry **160** and local media controller **135** to latch incoming commands.

(36) A controller (e.g., the local media controller **135** internal to the memory device **130**) controls access to the array of memory cells **104** in response to the commands and generates status information for the external memory sub-system controller **115**, i.e., the local media controller **135** is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells **104**. The local media controller **135** is in communication with row decode circuitry **108** and column decode circuitry **109** to control the row decode circuitry **108** and column decode circuitry **109** in response to the addresses. In one embodiment, local media controller **135** can perform a double single level cell (SLC) program operation to concurrently (i.e., at least partially overlapping in time) program memory cells in two or more separate sub-blocks of a block of memory array **104** using a single programming pulse.

(37) The local media controller **135** is also in communication with a cache register **172**. Cache register **172** latches data, either incoming or outgoing, as directed by the local media controller **135** to temporarily store data while the array of memory cells **104** is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data may be passed from the cache register **172** to the data register **170** for transfer to the array of memory cells **104**; then new data may be latched in the cache register **172** from the I/O control circuitry **160**. During a read operation, data may be passed from the cache register **172** to the I/O control circuitry **160** for output to the memory sub-system controller **115**; then new data may be passed from the data register **170** to the cache register **172**. The cache register **172** and/or the data register **170** may form

(e.g., may form a portion of) a page buffer of the memory device **130**. A page buffer may further include sensing devices (not shown in FIG. **1B**) to sense a data state of a memory cell of the array of memory cells **104**, e.g., by sensing a state of a data line connected to that memory cell. A status register **122** may be in communication with I/O control circuitry **160** and the local memory controller **135** to latch the status information for output to the memory sub-system controller **115**.

(38) Memory device **130** receives control signals at the memory sub-system controller **115** from the local media controller **135** over a control link **132**. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) may be further received over control link **132** depending upon the nature of the memory device **130**. In one embodiment, memory device **130** receives command signals (which represent commands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller **115** over a multiplexed input/output (I/O) bus **134** and outputs data to the memory sub-system controller **115** over I/O bus **134**.

(39) For example, the commands may be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **160** and may then be written into command register **124**. The addresses may be received over input/output (I/O) pins [7:0] of I/O bus **134** at I/O control circuitry **160** and may then be written into address register **114**. The data may be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry **160** and then may be written into cache register **172**. The data may be subsequently written into data register **170** for programming the array of memory cells **104**.

(40) In an embodiment, cache register **172** may be omitted, and the data may be written directly into data register **170**. Data may also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference may be made to I/O pins, they may include any conductive node providing for electrical connection to the memory device **130** by an external device (e.g., the memory sub-system controller **115**), such as conductive pads or conductive bumps as are commonly used.

(41) It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device **130** of FIG. **1B** has been simplified. It should be recognized that the functionality of the various block components described with reference to FIG. **1B** may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIG. **1B**. Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIG. **1B**. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) may be used in the various embodiments.

(42) FIG. **2** is a schematic of portions of an array of memory cells **104**, such as a NAND memory array, as could be used in a memory of the type described with reference to FIG. **1B** according to an embodiment. Memory array **104** includes access lines, such as wordlines **202.sub.0** to **202.sub.N**, and data lines, such as bit lines **204.sub.0** to **204.sub.M**. The wordlines **202** can be connected to global access lines (e.g., global wordlines), not shown in FIG. **2**, in a many-to-one relationship. For some embodiments, memory array **104** can be formed over a semiconductor that, for example, can be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n-well.

(43) Memory array **104** can be arranged in rows (each corresponding to a wordline **202**) and columns (each corresponding to a bit line **204**). Each column can include a string of series-connected memory cells (e.g., non-volatile memory cells), such as one of NAND strings **206.sub.0**

to **206.sub.M**. Each NAND string **206** can be connected (e.g., selectively connected) to a common source (SRC) **216** and can include memory cells **208.sub.0** to **208.sub.N**. The memory cells **208** can represent non-volatile memory cells for storage of data. The memory cells **208** of each NAND string **206** can be connected in series between a select gate **210** (e.g., a field-effect transistor), such as one of the select gates **210.sub.0** to **210.sub.M** (e.g., that can be source select transistors, commonly referred to as select gate source), and a select gate **212** (e.g., a field-effect transistor), such as one of the select gates **212.sub.0** to **212.sub.M** (e.g., that can be drain select transistors, commonly referred to as select gate drain). Select gates **210.sub.0** to **210.sub.M** can be commonly connected to a select line **214**, such as a source select line (SGS), and select gates **212.sub.0** to **212.sub.M** can be commonly connected to a select line **215**, such as a drain select line (SGD). Although depicted as traditional field-effect transistors, the select gates **210** and **212** can utilize a structure similar to (e.g., the same as) the memory cells **208**. The select gates **210** and **212** can represent a number of select gates connected in series, with each select gate in series configured to receive a same or independent control signal.

(44) A source of each select gate **210** can be connected to common source **216**. The drain of each select gate **210** can be connected to a memory cell **208.sub.0** of the corresponding NAND string **206**. For example, the drain of select gate **210.sub.0** can be connected to memory cell **208.sub.0** of the corresponding NAND string **206.sub.0**. Therefore, each select gate **210** can be configured to selectively connect a corresponding NAND string **206** to the common source **216**. A control gate of each select gate **210** can be connected to the select line **214**.

(45) The drain of each select gate **212** can be connected to the bit line **204** for the corresponding NAND string **206**. For example, the drain of select gate **212.sub.0** can be connected to the bit line **204.sub.0** for the corresponding NAND string **206.sub.0**. The source of each select gate **212** can be connected to a memory cell **208.sub.N** of the corresponding NAND string **206**. For example, the source of select gate **212.sub.0** can be connected to memory cell **208.sub.N** of the corresponding NAND string **206.sub.0**. Therefore, each select gate **212** can be configured to selectively connect a corresponding NAND string **206** to the corresponding bit line **204**. A control gate of each select gate **212** can be connected to select line **215**.

(46) The memory array **104** in FIG. 2 can be a quasi-two-dimensional memory array and can have a generally planar structure, e.g., where the common source **216**, NAND strings **206** and bit lines **204** extend in substantially parallel planes. Alternatively, the memory array **104** in FIG. 2 can be a three-dimensional memory array, e.g., where NAND strings **206** can extend substantially perpendicular to a plane containing the common source **216** and to a plane containing the bit lines **204** that can be substantially parallel to the plane containing the common source **216**.

(47) Typical construction of memory cells **208** includes a data-storage structure **234** (e.g., a floating gate, charge trap, and the like) that can determine a data state of the memory cell (e.g., through changes in threshold voltage), and a control gate **236**, as shown in FIG. 2. The data-storage structure **234** can include both conductive and dielectric structures while the control gate **236** is generally formed of one or more conductive materials. In some cases, memory cells **208** can further have a defined source/drain (e.g., source) **230** and a defined source/drain (e.g., drain) **232**. The memory cells **208** have their control gates **236** connected to (and in some cases form) a wordline **202**.

(48) A column of the memory cells **208** can be a NAND string **206** or a number of NAND strings **206** selectively connected to a given bit line **204**. A row of the memory cells **208** can be memory cells **208** commonly connected to a given wordline **202**. A row of memory cells **208** can, but need not, include all the memory cells **208** commonly connected to a given wordline **202**. Rows of the memory cells **208** can often be divided into one or more groups of physical pages of memory cells **208**, and physical pages of the memory cells **208** often include every other memory cell **208** commonly connected to a given wordline **202**. For example, the memory cells **208** commonly connected to wordline **202.sub.N** and selectively connected to even bit lines **204** (e.g., bit lines

204.sub.0, 204.sub.2, 204.sub.4, etc.) can be one physical page of the memory cells 208 (e.g., even memory cells) while memory cells 208 commonly connected to wordline 202.sub.N and selectively connected to odd bit lines 204 (e.g., bit lines 204.sub.1, 204.sub.3, 204.sub.5, etc.) can be another physical page of the memory cells 208 (e.g., odd memory cells).

(49) Although bit lines 204.sub.3-204.sub.5 are not explicitly depicted in FIG. 2, it is apparent from the figure that the bit lines 204 of the array of memory cells 104 can be numbered consecutively from bit line 204.sub.0 to bit line 204.sub.M. Other groupings of the memory cells 208 commonly connected to a given wordline 202 can also define a physical page of memory cells 208. For certain memory devices, all memory cells commonly connected to a given wordline can be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) can be deemed a logical page of memory cells. A block of memory cells can include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines 202.sub.0-202.sub.N (e.g., all NAND strings 206 sharing common wordlines 202). Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells. Although the example of FIG. 2 is discussed in conjunction with NAND flash, the embodiments and concepts described herein are not limited to a particular array architecture or structure, and can include other structures (e.g., SONOS, phase change, ferroelectric, etc.) and other architectures (e.g., AND arrays, NOR arrays, etc.).

(50) FIG. 3 is a schematic of portions of an array of memory cells implementing double single level cell (SLC) programming in accordance with some embodiments of the present disclosure. The portion of the array of memory cells, such as memory array 104, can be a block 300, for example. In one embodiment, the block 300 includes strings of memory cells that can be grouped into sub-blocks, such as sub-blocks 305.sub.0-305.sub.3. Other numbers of sub-blocks can be included in other embodiments.

(51) Specifically, in at least some embodiments, the block 300 includes a bit line 304, where each sub-block is coupled to the bit line 304. The first sub-block 305.sub.0 can include a first drain select (SGD) transistor 312.sub.0, a first source select (SGS) transistor 310.sub.0, and a first string of memory cells 306.sub.0 coupled therebetween. The second sub-block 305.sub.1 can include a second SGD transistor 312.sub.1, a second SGS transistor 310.sub.1, and a second string of memory cells 306.sub.1 coupled therebetween. The third sub-block 305.sub.2 can include a third SGD transistor 312.sub.2, a third SGS transistor 310.sub.2, and a third string of memory cells 306.sub.2 coupled therebetween. The fourth sub-block 305.sub.3 can include a fourth SGD transistor 312.sub.3, a fourth SGS transistor 310.sub.3, and a fourth string of memory cells 306.sub.3 coupled therebetween. By way of example, the first string of memory cells 306.sub.0 includes multiple memory cells 308.sub.0 . . . 308.sub.N. Each SGS transistor can be connected to a common source (SRC), such as a source voltage line, to provide voltage to the sources of the multiple memory cells 308.sub.0 . . . 308.sub.N. In some embodiments, the source voltage line includes a source plate that supplies the source voltage. In at least some embodiments, multiple wordlines (WLs) are coupled with gates of memory cells of each string of memory cells 306.sub.0 . . . 306.sub.3.

(52) In these embodiments, a first drain select gate line (SGD0) can be connected to the gate of the first SGD transistor 312.sub.0, a second drain select gate line (SGD1) can be connected to the gate of the second SGD transistor 312.sub.1, a third drain select gate line (SGD2) can be connected to the gate of the third SGD transistor 312.sub.2, and a fourth drain select gate line (SGD3) can be connected to the gate of the fourth SGD transistor 312.sub.3. Further, a first source select gate line (SGS0) can be connected to the gate of the first SGS transistor 310.sub.0, a second source select gate line (SGS1) can be connected to the gate of the second SGS transistor 310.sub.1, a third source select gate line (SGS2) can be connected to the gate of the third SGS transistor 310.sub.2,

and a fourth source select gate line (SGS3) can be connected to the gate of the fourth SGS transistor **310.sub.3**.

(53) In one embodiment, local media controller **135** can perform a double single level cell (SLC) program operation to concurrently program memory cells in two or more separate sub-blocks of block **300** using a single programming pulse applied to a selected wordline (e.g., WL.sub.N). In one embodiment, as part of a programming operation, the control logic causes a pass voltage (Vpass) to be applied to each of wordlines WL.sub.0-WL.sub.N concurrently. The pass voltage boosts a memory pillar channel voltage (e.g., due to gate-to-channel capacitive coupling) in each of sub-blocks **305.sub.0-305.sub.3** to a higher boost voltage (Vboost) during this phase of the programming operation. For example, as illustrated in FIG. **4A**, during a time period at the start of the programming operation known as tpass **402**, local media controller **135** can cause a voltage applied to all wordlines, including the selected WL.sub.N, to be ramped up to the pass voltage Vpass.

(54) Referring again to FIG. **3**, once each pillar channel voltage is boosted, local media controller **135** can selectively discharge the pillars of one or more sub-blocks according to a data pattern of bits to be programmed to block **300** during the program operation. In one embodiment, the data pattern is read from a storage location, such as a page buffer in memory device **130**, and represented by a voltage on the bitline **304**. For example, if a memory cell **309.sub.N** associated with the selected wordline WL.sub.N and located in sub-block **305.sub.1** is to be programmed, local media controller can activate the second SGD transistor **312.sub.1** by asserting a signal on the second drain select gate line (SGD1) to allow the boost voltage to be discharged onto the bitline **304**, thereby bringing the channel voltage in sub-block **305.sub.1** back to a ground voltage (e.g., 0V). Conversely, if a memory cell **308.sub.N** associated with the selected wordline WL.sub.N and located in sub-block **305.sub.0** is not to be programmed, local memory controller **135** will not activate the first SGD transistor **312.sub.0**, thereby causing the pillar channel voltage in sub-block **305.sub.0** to remain at the boost voltage. This sequence can be repeated for two or more sub-blocks, including for example sub-blocks **305.sub.2-305.sub.3** in addition. As illustrated in FIG. **4A**, during a subsequent time period **404**, the signals on the first and second drain select gate lines SGD0 and SGD1 are sequentially driven high to activate corresponding SGD transistors **312.sub.1** and **312.sub.0** and charge or discharge the corresponding pillar channels according to the data pattern represented on a bitline (BL), such as bitline **304**. In other embodiments, the order in which the signals on the first and second drain select gate lines SGD0 and SGD1 are driven high can change, such that SGD0 is driven high before SGD1, for example.

(55) Referring again to FIG. **3**, once complete, local media controller **135** can cause a single programming pulse (Vpgm) to be applied to the selected wordline WL.sub.N. Since the channel voltage in sub-block **305.sub.1** is at the ground voltage, the gate to channel voltage differential (e.g., Vpgm-GND) is large enough that memory cell **309.sub.N** will be programmed. Since the channel voltage in sub-block **305.sub.0** is at the boost voltage, the gate to channel voltage differential (e.g., Vpgm-Vboost) is too small, such that memory cell **308.sub.N** will not be programmed. As illustrated in FIG. **4A**, during a time period **406**, the signal on selected wordline WL.sub.N is ramped up to the program voltage Vpgm to program memory cells **308.sub.N** and **309.sub.N** according to the data pattern. Local media controller **135** can then perform either a double verify operation or a seamless verify operation during the subsequent program verify phase to confirm that memory cells **308.sub.N** and **309.sub.N** were properly programmed.

(56) FIG. **4B** illustrates an embodiment where the channel potential of at least one sub-block is increased (e.g., to the boost voltage) according to the data pattern prior to the pass voltage being applied. As illustrated in FIG. **4B**, during an initial time period **410**, the signal on the first drain select gate lines SGD0 is driven high to activate corresponding SGD transistor **312.sub.0** allowing the voltage representing the data pattern on the bitline (BL) to charge the corresponding pillar channel for sub-block **310.sub.0**. In one embodiment, the initial time period **410** overlaps with a

seeding phase of the program operation. Then, during the time period tpass **412**, local media controller **135** can cause the voltage applied to all wordlines, including the selected WL.sub.N, to be ramped up to the pass voltage Vpass. During the subsequent time period **414**, the signal on the second drain select gate line SGD1 is driven high to activate corresponding SGD transistor **312.sub.1** and charge or discharge the corresponding pillar channel according to the data pattern represented on the bitline (BL). During time period **416**, the signal on selected wordline WL.sub.N is ramped up to the program voltage Vpgm to program memory cells **308.sub.N** and **309.sub.N** according to the data pattern. In one embodiment, the signal SGD1 remains in a high state during time period **416** as well. Local media controller **135** can then perform either a double verify operation or a seamless verify operation during the subsequent program verify phase to confirm that memory cells **308.sub.N** and **309.sub.N** were properly programmed.

(57) FIG. **4C** illustrates an embodiment where the channel potential of at least one sub-block is increased (e.g., to the boost voltage) according to the data pattern prior to the pass voltage being applied. The embodiment illustrated in FIG. **4C**, is largely the same as that illustrated in FIG. **4B**, except that in FIG. **4C**, after the signal SGD1 is driven high during time period **424** to activate corresponding SGD transistor **312.sub.1**, the signal SGD1 is driven low rather than remaining in the high state while the programming pulse is applied on the selected WL.sub.N.

(58) FIG. **5** is a flow diagram of an example method of double single level cell (SLC) programming in a memory device of a memory sub-system in accordance with some embodiments of the present disclosure. The method **500** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **500** is performed by local media controller **135** of FIG. **1A** and FIG. **1B**. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

(59) At operation **505**, a program operation is initiated. In one embodiment, the program operation includes a program phase and a program verify phase. In certain embodiments, each of these phases can be repeated numerous times in a cycle during a single program operation. During the program phase, a program voltage is applied to selected wordlines of the memory device **130**, in order to program a certain level of charge to the selected memory cells on the wordlines representative of a desired value. The desired value can be represented by a multi-bit data pattern (e.g., stored in a page buffer of the memory device **130**), where each bit is to be stored in a separate memory cell associated with the selected wordline.

(60) At operation **510**, the channel potential of a sub-block of a block of memory array **104** of memory device **130** is optionally increased. For example, control logic (e.g., local media controller **135**) can cause the channel potential of at least one of sub-blocks **305.sub.0-305.sub.3** to be increased to a boost voltage (Vboost) according to a data pattern prior to causing a pass voltage to be applied to the wordlines of the block. In one embodiment, the signal on one drain select gate line, such as the first drain select gate line SGD0, is driven high to activate a corresponding SGD transistor, such as SGD transistor **312.sub.0**, allowing the voltage representing the data pattern on the bitline (BL) to charge the corresponding pillar channel for sub-block **305.sub.0**, as described above with respect to FIG. **4B**.

(61) At operation **515**, a pass voltage is applied to the wordlines of a block of memory array **104** of memory device **130**. For example, control logic (e.g., local media controller **135**) can cause the pass voltage (Vpass) to be applied to a plurality of wordlines of the block, such as each of wordlines WL.sub.0-WL.sub.N, concurrently. In one embodiment, the block, such as block **300**,

includes sub-blocks **305.sub.0-305.sub.3**, each including a string of memory cells surrounding a pillar of channel material. The pass voltage boosts a memory pillar channel voltage (e.g., due to gate to channel capacitive coupling) in each of sub-blocks **305.sub.0-305.sub.3** to a higher boost voltage (Vboost) during this phase of the program operation, although sub-block **305.sub.0** may already have a pillar channel potential at the boost voltage if optionally pre-boosted at operation **510**.

(62) At operation **520**, one or more sub-blocks are selectively discharged. For example, the control logic can selectively discharge the boost voltage from one or more of the sub-blocks **305.sub.0-305.sub.3** according to the data pattern representing a sequence of bits to be programmed to respective memory cells of the plurality of sub-blocks. In one embodiment, the data pattern is read from a storage location, such as a page buffer in memory device **130**, and represented by a voltage on the bitline **304**. For example, if a memory cell **309.sub.N** associated with the selected wordline WL.sub.N and located in sub-block **305.sub.1** is to be programmed, local media controller can activate the second SGD transistor **312.sub.1** by asserting a signal on the second drain select gate line (SGD1) to allow the boost voltage to be discharged onto the bitline **304**, thereby bringing the channel voltage in sub-block **305.sub.1** back to a ground voltage (e.g., 0V). Conversely, if a memory cell **308.sub.N** associated with the selected wordline WL.sub.N and located in sub-block **305.sub.0** is not to be programmed, local memory controller **135** will not activate the first SGD transistor **312.sub.0**, thereby causing the pillar channel voltage in sub-block **305.sub.0** to remain at the boost voltage. This sequence can be repeated for two or more sub-blocks, including for example sub-blocks **305.sub.2-305.sub.3** in addition, although need not be performed for any sub-blocks, such as sub-block **305.sub.0**, that were optionally pre-boosted at operation **510**.

(63) At operation **525**, a programming pulse is applied to the selected wordline. For example, the control logic can cause a single programming pulse to be applied to the selected wordline WL.sub.N of the plurality of wordlines of the block **300** to program the respective memory cells of the plurality of sub-blocks according to the data pattern. Since the channel voltage in sub-block **305.sub.1** is at the ground voltage, the gate to channel voltage differential (e.g., Vpgm-GND) is large enough that memory cell **309.sub.N** will be programmed. Since the channel voltage in sub-block **305.sub.0** is at the boost voltage, the gate to channel voltage differential (e.g., Vpgm-Vboost) is too small, such that memory cell **308.sub.N** will not be programmed.

(64) At operation **530**, a program verify phase is initiated. During the program verify phase, a read voltage is applied to the selected word lines to read the level of charge stored at the selected memory cells to confirm that the desired value was properly programmed. Depending on the implementation, certain memory devices can utilize either a double verify operation or a seamless verify operation during the program verify phase.

(65) FIG. **6** illustrates an example machine of a computer system **600** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system **600** can correspond to a host system (e.g., the host system **120** of FIG. **1**) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory sub-system **110** of FIG. **1**) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to the local media controller **135** of FIG. **1**). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

(66) The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term

“machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(67) The example computer system **600** includes a processing device **602**, a main memory **604** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **606** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system **618**, which communicate with each other via a bus **630**.

(68) Processing device **602** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **602** can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **602** is configured to execute instructions **626** for performing the operations and steps discussed herein. The computer system **600** can further include a network interface device **608** to communicate over the network **620**.

(69) The data storage system **618** can include a machine-readable storage medium **624** (also known as a computer-readable medium) on which is stored one or more sets of instructions **626** or software embodying any one or more of the methodologies or functions described herein. The instructions **626** can also reside, completely or at least partially, within the main memory **604** and/or within the processing device **602** during execution thereof by the computer system **600**, the main memory **604** and the processing device **602** also constituting machine-readable storage media. The machine-readable storage medium **624**, data storage system **618**, and/or main memory **604** can correspond to the memory sub-system **110** of FIG. 1.

(70) In one embodiment, the instructions **626** include instructions to implement functionality corresponding to the local media controller **135** of FIG. 1). While the machine-readable storage medium **624** is shown in an example embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

(71) Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

(72) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or

similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage systems.

(73) The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

(74) The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

(75) The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). In some embodiments, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory components, etc.

(76) In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A memory device comprising: a memory array; and control logic, operatively coupled with the memory array, to perform operations comprising: prior to applying a pass voltage to a plurality of wordlines of a block of the memory array, causing a channel potential of one sub-block of a plurality of sub-blocks of the block of the memory array to be increased to a boost voltage according to a data pattern representing a sequence of bits to be programmed to respective memory cells of the plurality of sub-blocks; causing the pass voltage to be applied to the plurality of wordlines of the block of the memory array, the pass voltage to boost a channel potential of the plurality of sub-blocks to a higher boost voltage; selectively discharging the higher boost voltage from one or more of the plurality of sub-blocks according to the data pattern; and causing a single programming pulse to be applied to a selected wordline of the plurality of wordlines of the block to program the respective memory cells of the plurality of sub-blocks according to the data pattern during a program operation.

2. The memory device of claim 1, wherein each wordline of the plurality of wordlines is coupled to a respective set of memory cells in the block, and wherein one memory cell from each respective set of memory cells is associated with one of the plurality of sub-blocks of the block.

3. The memory device of claim 2, wherein each of the plurality of sub-blocks comprises a string of memory cells sharing a pillar of channel material, and wherein each memory cell in the string of memory cells is associated with a respective wordline of the plurality of wordlines.
4. The memory device of claim 3, wherein the plurality of sub-blocks comprises respective select gate devices to couple the strings of memory cells to a bit line.
5. The memory device of claim 4, wherein selectively discharging the higher boost voltage from one or more of the plurality of sub-blocks according to the data pattern comprises: activating one or more of the respective select gate devices corresponding to one or more first sub-blocks for which the respective memory cells are to be programmed to cause the channel potential of the one or more first sub-blocks to decrease to a ground potential; and not activating one or more of the respective select gate devices corresponding to one or more second sub-blocks for which the respective memory cells are not to be programmed to cause the channel potential of the one or more second sub-blocks to remain at the higher boost voltage.
6. The memory device of claim 5, wherein the single programming pulse applied to the selected wordline of the plurality of wordlines of the block is to program the respective memory cells of the one or more first sub-blocks and not program the respective memory cells of the one or more second sub-blocks.
7. The memory device of claim 6, wherein the one or more of the respective select gate devices corresponding to the one or more first sub-blocks remain activated while the single programming pulse is applied to the selected wordline.
8. The memory device of claim 1, wherein causing the channel potential of the at least one sub-block of a plurality of sub-blocks to be increased to the boost voltage is performed concurrently with a seeding phase of the program operation.
9. A method comprising: prior to applying a pass voltage to a plurality of wordlines of a block of a memory array of a memory device, causing a channel potential of one sub-block of a plurality of sub-blocks of the block of the memory array to be increased to a boost voltage according to a data pattern representing a sequence of bits to be programmed to respective memory cells of the plurality of sub-blocks; causing the pass voltage to be applied to the plurality of wordlines of the block of the memory array, the pass voltage to boost a channel potential of the plurality of sub-blocks to a higher boost voltage; selectively discharging the higher boost voltage from one or more of the plurality of sub-blocks according to the data pattern; and causing a single programming pulse to be applied to a selected wordline of the plurality of wordlines of the block to program the respective memory cells of the plurality of sub-blocks according to the data pattern during a program operation.
10. The method of claim 9, wherein each wordline of the plurality of wordlines is coupled to a respective set of memory cells in the block, and wherein one memory cell from each respective set of memory cells is associated with one of the plurality of sub-blocks of the block.
11. The method of claim 10, wherein each of the plurality of sub-blocks comprises a string of memory cells sharing a pillar of channel material, and wherein each memory cell in the string of memory cells is associated with a respective wordline of the plurality of wordlines.
12. The method of claim 11, wherein the plurality of sub-blocks comprises respective select gate devices to couple the strings of memory cells to a bit line.
13. The method of claim 12, wherein selectively discharging the higher boost voltage from one or more of the plurality of sub-blocks according to the data pattern comprises: activating one or more of the respective select gate devices corresponding to one or more first sub-blocks for which the respective memory cells are to be programmed to cause the channel potential of the one or more first sub-blocks to decrease to a ground potential; and not activating one or more of the respective select gate devices corresponding to one or more second sub-blocks for which the respective memory cells are not to be programmed to cause the channel potential of the one or more second sub-blocks to remain at the higher boost voltage.

14. The method of claim 13, wherein the single programming pulse applied to the selected wordline of the plurality of wordlines of the block is to program the respective memory cells of the one or more first sub-blocks and not program the respective memory cells of the one or more second sub-blocks.

15. The method of claim 14, wherein the one or more of the respective select gate devices corresponding to the one or more first sub-blocks remain activated while the single programming pulse is applied to the selected wordline.

16. The method of claim 9, wherein causing the channel potential of the at least one sub-block of a plurality of sub-blocks to be increased to the boost voltage is performed concurrently with a seeding phase of the program operation.
